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1. Introduction

GNN4EEG is an open benchmark and software toolkit for graph-neural-network-based electroencephalography classification. It addresses a central limitation of conventional EEG classifiers: flattening multi-channel signals can weaken or remove the spatial relationships among scalp electrodes. GNN4EEG instead represents each EEG channel as a graph node and learns or constructs channel relationships as graph edges.

The benchmark is built on the FACED affective-computing dataset, which contains recordings from 123 participants watching 28 emotion-elicitation video clips. The paper defines four tasks—cross-9, cross-2, intra-9, and intra-2—together with four models (DGCNN, RGNN, SparseDGCNN, and HetEmotionNet) and three validation protocols (CV, FCV, and NCV).

This report evaluates the uploaded GNN4EEG paper and two cross-9 notebooks. The first notebook implements the DE/LDS cross-9 experiment for DGCNN, RGNN, and SparseDGCNN. The second notebook completes a ten-fold HetEmotionNet-style experiment from subject-wise processed feature files. The report follows the organization and evidence discipline of the provided reproduction-report template.

2. Why GNN4EEG Was Selected

Selection criterion	Justification
Dataset scale	FACED includes 123 participants, substantially larger than several common affective-EEG benchmarks.
Graph suitability	EEG electrodes naturally form nodes, while anatomical, statistical, or learned relations form edges.
Task relevance	Cross-9 tests subject-independent nine-emotion classification, the most difficult reported benchmark setting.
Model breadth	The toolkit compares learnable, signed, sparse, and heterogeneous graph architectures.
Evaluation value	CV, fixed-epoch CV, and nested CV expose how validation design changes the reported result.
Reproduction value	The uploaded notebooks contain explicit data checks, leakage checks, hyperparameter selection, checkpointing, and resume logic.

3. GNN4EEG Framework and Model Architecture

GNN4EEG cross-9 reproduction workflow

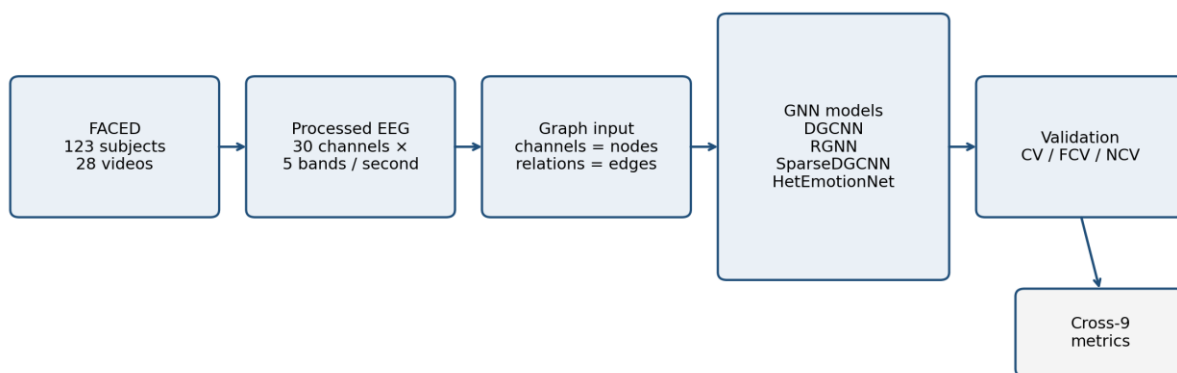


Fig. 1. GNN4EEG cross-9 reproduction workflow used in the uploaded notebooks.

3.1 Benchmark construction and preprocessing

The paper retains the final 30 seconds of every video, removes the last two channels, and represents each one-second window by five differential-entropy features: delta, theta, alpha, beta, and gamma. The resulting benchmark representation contains 30 EEG nodes and five node features per sample. For the nine-class task, neutral contains four videos and each other emotion contains three videos.

Benchmark element	Paper specification
Participants	123
Videos per participant	28
Retained duration	30 seconds per video
Graph nodes	30 EEG channels after removing the final two channels
Node features	Five DE bands smoothed by linear dynamic systems
Cross-9 classes	Anger, Disgust, Fear, Sadness, Neutral, Amusement, Inspiration, Joy, Tenderness
One-second samples	$123 \times 28 \times 30 = 103,320$

3.2 Graph representation

For an EEG graph $G = (V, E)$, V is the set of channels and E represents channel relationships. With node features $X \in \mathbb{R}^{(n \times d)}$, weighted adjacency A , and degree matrix D , one graph-convolution layer is expressed as:

$$H^{(l+1)} = \sigma(D^{(-1/2)} A D^{(-1/2)} H^{(l)} W^{(l)})$$

The models differ mainly in how A is initialized, learned, constrained, and combined with temporal processing.

3.3 DGCNN

DGCNN learns a complete symmetric adjacency matrix during training. In the uploaded main notebook, negative edge weights are projected to zero after every optimizer step. Two SGConv-style propagation steps map five input features to the selected hidden dimension; all node embeddings are then flattened and classified through a 64-unit fully connected layer.

3.4 RGNN

RGNN begins with a distance-based adjacency derived from the 30 electrode coordinates and introduces selected negative global connections. It uses signed adjacency normalization and graph-level sum pooling. For the cross-subject task, the notebook enables NodeDAT domain-adversarial training through gradient reversal; test features remain untouched until final evaluation.

3.5 SparseDGCNN

SparseDGCNN retains the DGCNN classifier but applies proximal soft-thresholding to the learned adjacency after each optimizer step. This explicitly promotes edge sparsity rather than merely adding an L1 term to the loss.

3.6 HetEmotionNet

The paper describes HetEmotionNet as a two-stream heterogeneous graph recurrent network that captures spatial-spectral and spatial-temporal information using graph transformation and gated recurrent processing. The uploaded HetEmotionNet notebook implements an EEG-only adaptation: a frequency stream of shape 30×5 and a temporal stream of shape 30×30 each undergo normalized graph propagation and a bidirectional GRU, followed by projection, flattening, fusion, and nine-class classification.

3.7 Validation protocols

Protocol	Selection and reporting rule	Reliability implication
CV	Each fold reports its best validation epoch; results are averaged.	Optimistic because each fold may select a different best epoch.
FCV	One epoch is selected from the average fold curve and used for all folds.	Reduces fold-specific epoch selection but still evaluates on validation data.
NCV	Inner folds select hyperparameters and epoch; an untouched outer fold is evaluated once.	Most defensible estimate of subject-independent generalization.

4. Original Paper Results on FACED

The paper reports the following cross-9 accuracy values. NCV is the primary comparison because both uploaded notebooks are designed around nested cross-validation.

Model	CV (%)	FCV (%)	NCV (%)
DGCNN	30.8	28.9	26.7
RGNN	34.0	32.1	31.3
SparseDGCNN	30.7	29.0	27.7
HetEmotionNet	32.5	30.8	30.8

RGNN is the best published cross-9 model under all three protocols. The paper attributes its cross-subject advantage to NodeDAT. HetEmotionNet is strongest in the intra-subject settings, but its cross-9 NCV accuracy is 30.8%.

5. Reproduction Scope

Component	Reproduction target
Dataset	FACED processed features
Task	Cross-subject, nine-class emotion classification (cross-9)
Samples	One one-second graph sample for each retained second
Nodes and features	30 channels; five frequency-band features
Main notebook models	DGCNN, RGNN, SparseDGCNN
Second notebook model	EEG-only HetEmotionNet-style two-stream network
Validation	10 outer folds and 3 inner folds
Hyperparameter grid	Hidden dimensions 20/40/80; learning rates 1e-4/1e-3/1e-2
Primary metric	Outer-test accuracy; macro metrics reported where available
Evidence boundary	Completed numerical results are available only for the HetEmotionNet notebook

6. Exact Reproduction Requirements and Availability

Evidence boundary

The uploaded main cross-9 notebook contains complete code but no stored execution outputs. Consequently, DGCNN, RGNN, and SparseDGCNN results cannot be reconstructed or claimed from that file. Only the HetEmotionNet notebook contains verified ten-fold outputs.

Requirement	Availability	Reproduction decision
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Requirement	Availability	Reproduction decision
GNN4EEG paper and published table	Available	Used as the original benchmark source.
FACED DE/LDS MAT structure	Defined in main notebook	Canonical shape 123×840×150; no execution output is stored.
Subject-wise processed PKL files	Verified in Het notebook outputs	123 accepted files, canonicalized to 123×28×30×30×5.
Exact paper fold identities	Unavailable	Seed-42 subject permutations were used.
Exact original random seed	Unavailable	Seed 42 was fixed in both notebooks.
Official DGCNN/RGNN/Sparse outputs	Unavailable in uploaded notebook	Implementation status only; no result table reported.
Official HetEmotionNet sample-specific adjacency	Not reproduced exactly	One graph was calculated per 30-second clip and reused for its 30 samples.
Exact HetEmotionNet feature family	Ambiguous	The accepted-file manifest shown in output points to PSD paths, whereas the paper benchmark specifies DE+LDS.
Full original heterogeneous model	Not reproduced exactly	A standalone graph-BiGRU two-stream classifier was implemented.
Hardware match	Unavailable	Completed HetEmotionNet run used a Tesla T4; the GNN4EEG paper used an RTX 3090.

7. Notebook Design

7.1 Main cross-9 notebook

The main notebook is organized as a full, exact-NCV pipeline:

Imports → MAT discovery and robust loading → canonical shape conversion → nine-class labeling → graph reshaping → dataset quality checks → subject-wise normalization → deterministic outer and inner subject folds → DGCNN/RGNN/SparseDGCNN definitions → complete inner validation curves for all 100 epochs → outer retraining → test evaluation → checkpoint and result export.

The design is strong from a reproducibility-engineering perspective. It disables early stopping, stores every epoch, validates checkpoint metadata, preserves RNG and DataLoader states, supports CPU-safe restoration, and uses latest/previous checkpoint fallbacks.

7.2 HetEmotionNet resume notebook

The completed HetEmotionNet notebook follows this sequence:

Automatic workspace restoration → recursive PKL discovery → shape canonicalization and deduplication → atomic cache creation → subject-wise z-score normalization → frequency and temporal stream reconstruction → clip-level mutual-information adjacency → subject-disjoint nested folds → graph-BiGRU model → resumable inner-grid selection → fresh outer training → test predictions → aggregate metrics, confusion matrix, fold plot, sample inference, and checkpoint archives.

The resume mechanism is operationally important because the full design requires 10 outer folds × 9 hyperparameter combinations × 3 inner folds, followed by 10 final outer models. The notebook saves atomic last.pt checkpoints and creates rolling ZIP bundles for continuation across Kaggle sessions.

8. Experimental Configuration

8.1 Main notebook configuration

Setting	Value used	Source/status
Data file	FACED_dataset_9_labels.mat; key de_lds	Configured in notebook; no saved outputs
Canonical data	123×840×150	Paper-aligned representation
Model input	103,320×30×5	One-second graph samples

Setting	Value used	Source/status
Models	DGCNN, RGNN, SparseDGCNN	All enabled
Protocol	Cross-subject NCV	10 outer, 3 inner folds
Test subjects	12 in folds 1–9; 15 in fold 10	Paper-aligned fold sizes
Hidden dimensions	20, 40, 80	Full grid
Learning rates	0.0001, 0.001, 0.01	Full grid
Epoch candidates	1–100	No early stopping
GNN layers	2	Paper setting
Dropout	0.5	RGNN classifier
Batch size	256	Paper setting
L1 / L2	0.005 / 0.005	Cross-9 setting
Subject normalization	Per-subject z-score	Enabled
Random seed	42	Reproduction decision

8.2 HetEmotionNet configuration

Setting	Value used	Evidence
Accepted subjects	123	Notebook output
Canonical feature array	123×28×30×30×5	Notebook output after dropping two channels
One-second samples	103,320	Notebook output
Frequency stream	30×5 per sample	Current second across five bands
Temporal stream	30×30 per clip	30 seconds averaged across bands
Graph cache	3,444 matrices of 30×30	One per subject-video clip
Adjacency	Normalized mutual information, 10 bins, threshold 0.70	Notebook configuration
Adjacency density	4.47% non-zero entries	Notebook output
Model	Two graph-BiGRU streams + projection + fusion	Standalone adaptation
Hidden dimensions	20, 40, 80	Full NCV grid
Learning rates	0.0001, 0.001, 0.01	Full NCV grid
Maximum epochs	100	Inner curves
Optimizer	Adam	Notebook code
Loss	Cross-entropy + mean-normalized L1/L2	Notebook code
Dropout / batch size	0.5 / 256	Notebook configuration
Hardware	Tesla T4; CUDA verified	Notebook output
Software	Python 3.12.13; PyTorch 2.10.0+cu128	Notebook output

8.3 Why the HetEmotionNet result is not a strict replication

The completed HetEmotionNet experiment should be described as a GNN4EEG-inspired reimplementaion rather than an exact replication. Four differences are decisive:

1. The paper benchmark specifies DE features smoothed by LDS, while the accepted-file manifest displayed by the notebook points to PSD feature paths.

2. The original HetEmotionNet uses heterogeneous graph transformation, whereas the notebook implements normalized graph propagation followed by bidirectional GRUs.
3. The official approach is described as sample-specific; the notebook constructs one adjacency matrix per 30-second clip and reuses it for all 30 one-second samples.
4. Exact original subject assignments and random seed are unavailable, so the seed-42 folds are not identity-matched to the paper.

These differences are large enough that a higher notebook accuracy cannot be interpreted as an improvement over the published model.

9. Evaluation Metrics

Metric	Meaning	Use in this report
Accuracy	Correct predictions divided by all predictions.	Primary paper comparison.
Macro precision	Mean precision over the nine classes.	Treats each class equally.
Macro recall	Mean recall over the nine classes.	Shows average class sensitivity.
Macro-F1	Mean of the nine class-specific F1 scores.	Main balance-sensitive metric.
Weighted F1	F1 weighted by class support.	Defined in the main notebook, but no output is stored.
Per-class metrics	Precision, recall, and F1 for each emotion.	Identifies easy and difficult emotions.
Confusion matrix	Counts or row-normalized proportions of true versus predicted classes.	Shows the direction of class confusion.
Mean $\pm$ standard deviation	Average and variability over ten outer folds.	Measures cross-subject stability.

10. Reproduced Results

10.1 Main notebook execution status

No verifiable DGCNN/RGNN/SparseDGCNN result

The uploaded Resumed-Cross9_3rd.ipynb contains 90 cells and complete experiment code, but every code cell has an empty output list. No fold metrics, selected configurations, confusion matrices, or aggregate result files are embedded. Reporting numerical results for these three models would therefore be unsupported.

10.2 HetEmotionNet ten-fold results

Outer fold	Hidden	Learning rate	Selected epoch	Accuracy (%)	Macro-F1 (%)
1	40	0.0100	3	48.68	48.55
2	20	0.0100	8	44.30	43.53
3	40	0.0100	9	40.65	40.75
4	40	0.0100	45	48.40	48.30
5	80	0.0100	20	48.36	47.71
6	40	0.0100	7	40.75	40.87
7	40	0.0100	13	44.84	44.88
8	80	0.0100	3	44.18	44.18
9	40	0.0001	21	50.82	50.57
10	40	0.0100	5	48.56	48.11

Fold 9 achieved the highest accuracy (50.82%), while fold 3 produced the lowest (40.65%). The range is 10.17 percentage points, demonstrating meaningful variation in unseen-subject difficulty. Hidden dimension 40 was selected

in seven folds; learning rate 0.01 was selected in nine folds. The mean selected final-training epoch was 13.4, with a range of 3–45 epochs.

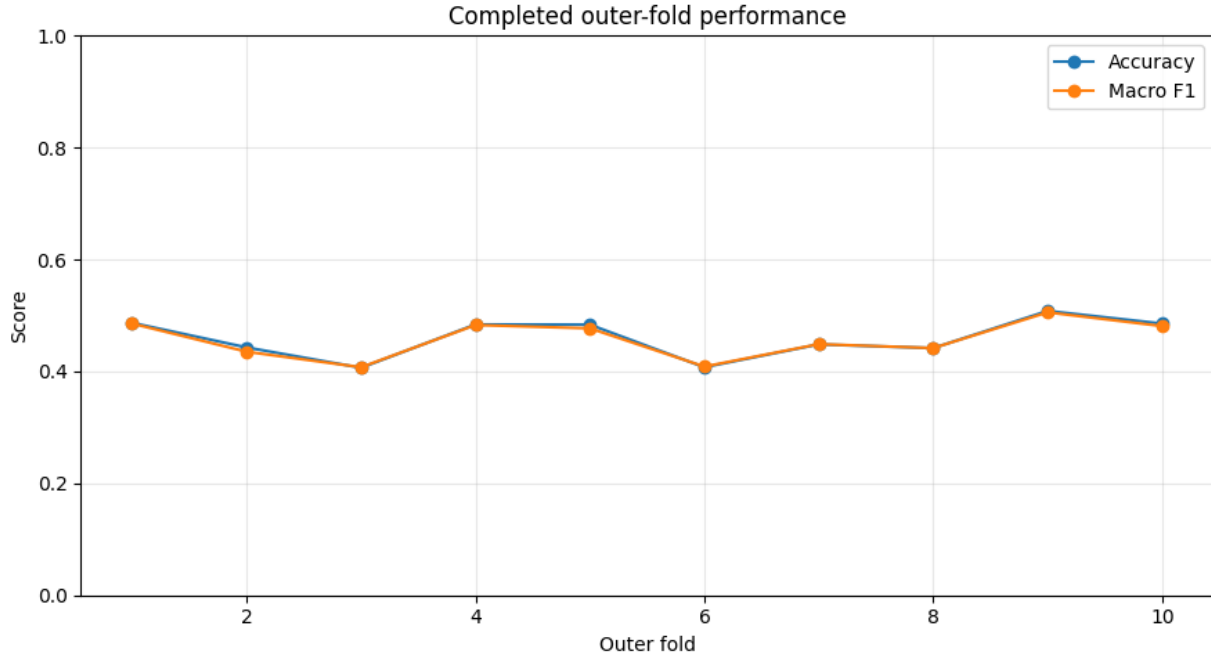


Fig. 2. Completed outer-fold accuracy and macro-F1 from the HetEmotionNet notebook.

10.3 Aggregate and per-class results

Metric	Mean $\pm$ standard deviation	Minimum	Maximum
Accuracy	45.95 $\pm$ 3.53%	40.65%	50.82%
Macro precision	46.90 $\pm$ 3.42%	41.22%	50.82%
Macro recall	46.21 $\pm$ 3.46%	40.78%	50.88%
Macro-F1	45.74 $\pm$ 3.40%	40.75%	50.57%

Because fold 10 contains 15 test subjects while the other folds contain 12, the pooled sample-level accuracy is 46.02%, slightly different from the unweighted mean of fold accuracies (45.95%).

Emotion	Precision	Recall	F1	Support
Anger	0.42	0.43	0.43	11,070
Disgust	0.44	0.43	0.44	11,070
Fear	0.50	0.50	0.50	11,070
Sadness	0.44	0.36	0.39	11,070
Neutral	0.43	0.39	0.41	14,760
Amusement	0.53	0.67	0.60	11,070
Inspiration	0.43	0.42	0.43	11,070
Joy	0.55	0.51	0.53	11,070
Tenderness	0.40	0.46	0.42	11,070

Amusement is the easiest class (F1 = 0.60 and recall = 0.67), followed by Joy (F1 = 0.53). Sadness is the most difficult class (F1 = 0.39), and neutral recall is also low (0.39) despite having the largest support.

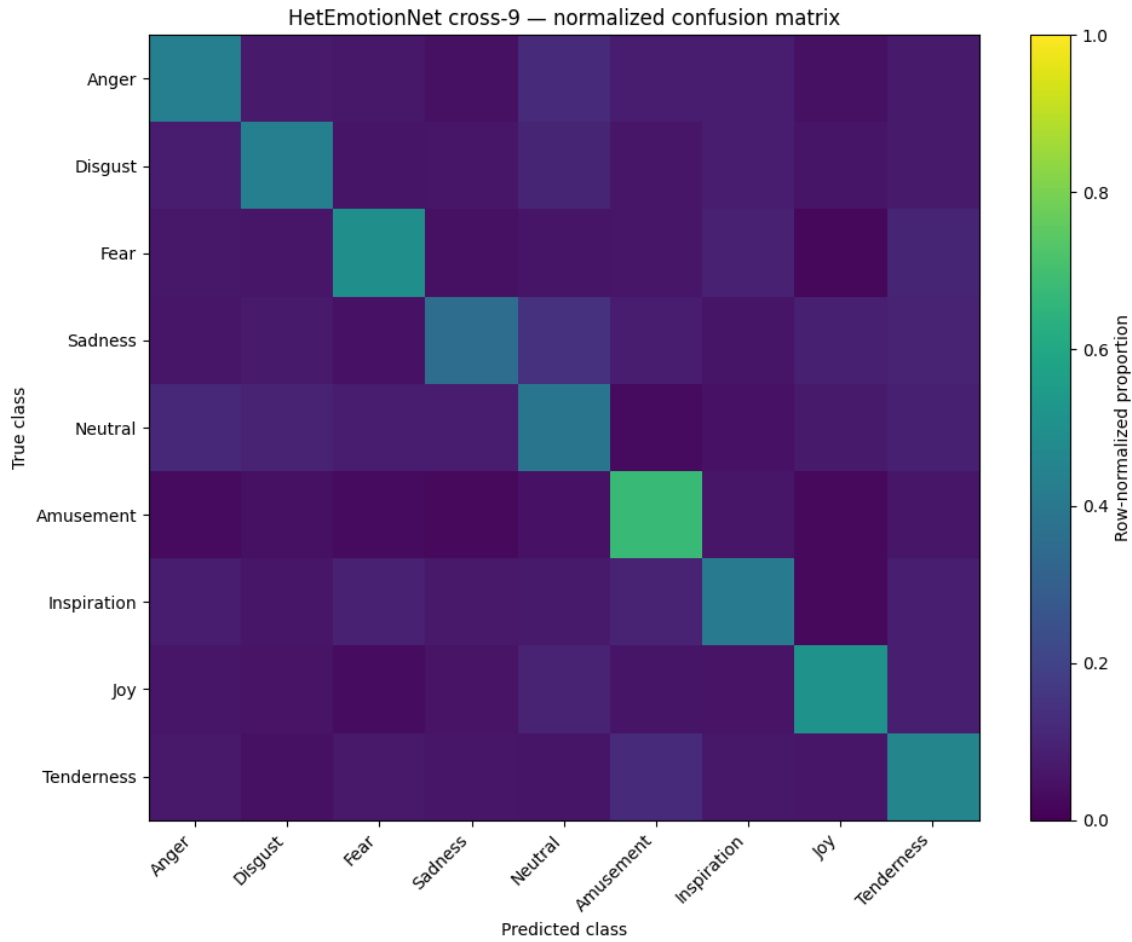


Fig. 3. Row-normalized pooled confusion matrix from all ten HetEmotionNet outer folds.

11. Original Versus Reproduced Results

Measure	Original GNN4EEG	Implemented model results	Difference	Comparison status
HetEmotionNet cross-9 NCV accuracy	30.8%	$45.95 \pm 3.53\%$	+15.15 percentage points	Not directly comparable
HetEmotionNet macro-F1	Not reported	$45.74 \pm 3.40\%$	—	Descriptive only
DGCNN cross-9 NCV accuracy	26.7%		—	Not reproduced in evidence
RGNN cross-9 NCV accuracy	31.3%		—	Not reproduced in evidence
SparseDGCNN cross-9 NCV accuracy	27.7%		—	Not reproduced in evidence
Test sample total	103,320	103,320	Same count	Comparable count only
Outer fold sizes	$12 \times 9 + 15 \times 1$ subjects	$12 \times 9 + 15 \times 1$ subjects	Same sizes	Exact identities differ

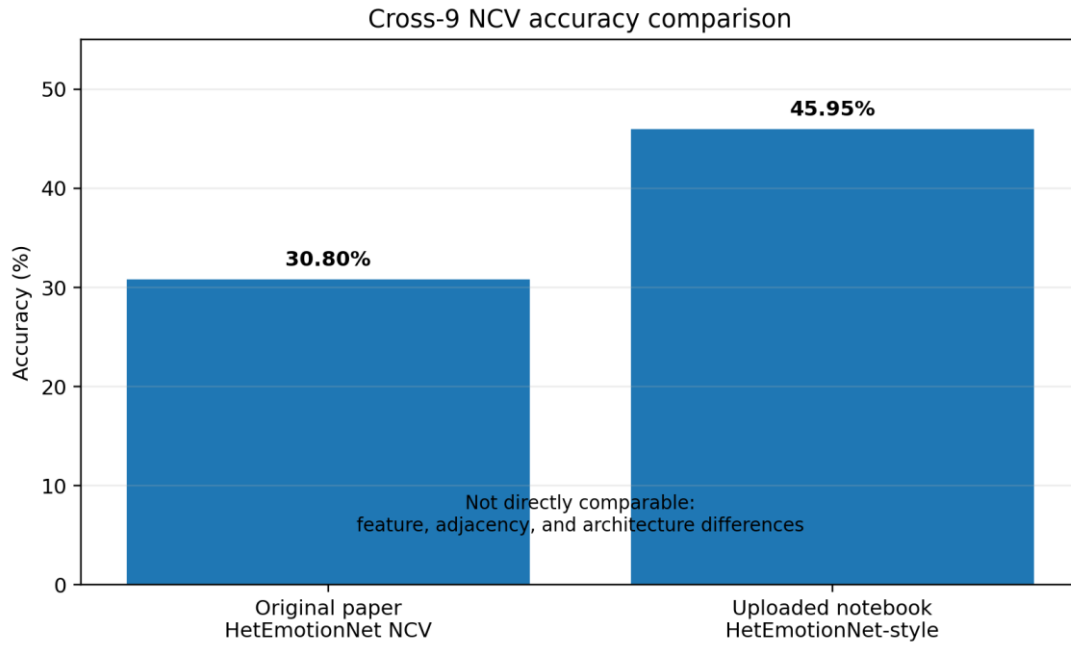


Fig. 4. Published and notebook HetEmotionNet accuracy; the values are not parameter- or pipeline-identical.

The +15.15-point difference is too large to attribute to random fold variation alone. The most plausible explanation is the combined change in feature source, graph construction, and HetEmotionNet architecture. It should not be presented as a successful exact replication or as evidence that the notebook model outperforms the published implementation.

12. Source and Implementation Differences

Difference	Paper / main source	Implemented HetEmotionNet	Possible effect
Feature family	DE features smoothed by LDS	Accepted manifest displays PSD paths	Changes the statistical input representation.
Input source	Consolidated benchmark representation	Subject-wise PKL files with duplicate selection	May select a different processed feature variant.
Het architecture	Heterogeneous graph transformation + recurrent modeling	Two normalized graph-BiGRU streams	Different capacity and inductive bias.
Adjacency frequency	Described as sample-specific in implementation	One graph per 30-second clip	All 30 clip samples share the same graph.
Adjacency sparsity	Not reported at the notebook threshold	4.47% non-zero globally; clip 0 is self-loops only	Some samples receive little inter-channel message passing.
Temporal input	Paper toolkit implementation details not fully exposed in article	Band-averaged 30-second clip reused by every second	Creates a stronger clip-level context feature.
Fold identities	Not listed	Seed-42 subject permutation	Different unseen-subject difficulty.
Software/hardware	RTX 3090 reported for toolkit experiments	Tesla T4, PyTorch 2.10.0	Numerical and runtime differences.
Main notebook outputs	Paper reports model results	Uploaded file has no stored outputs	Prevents verification of three models.

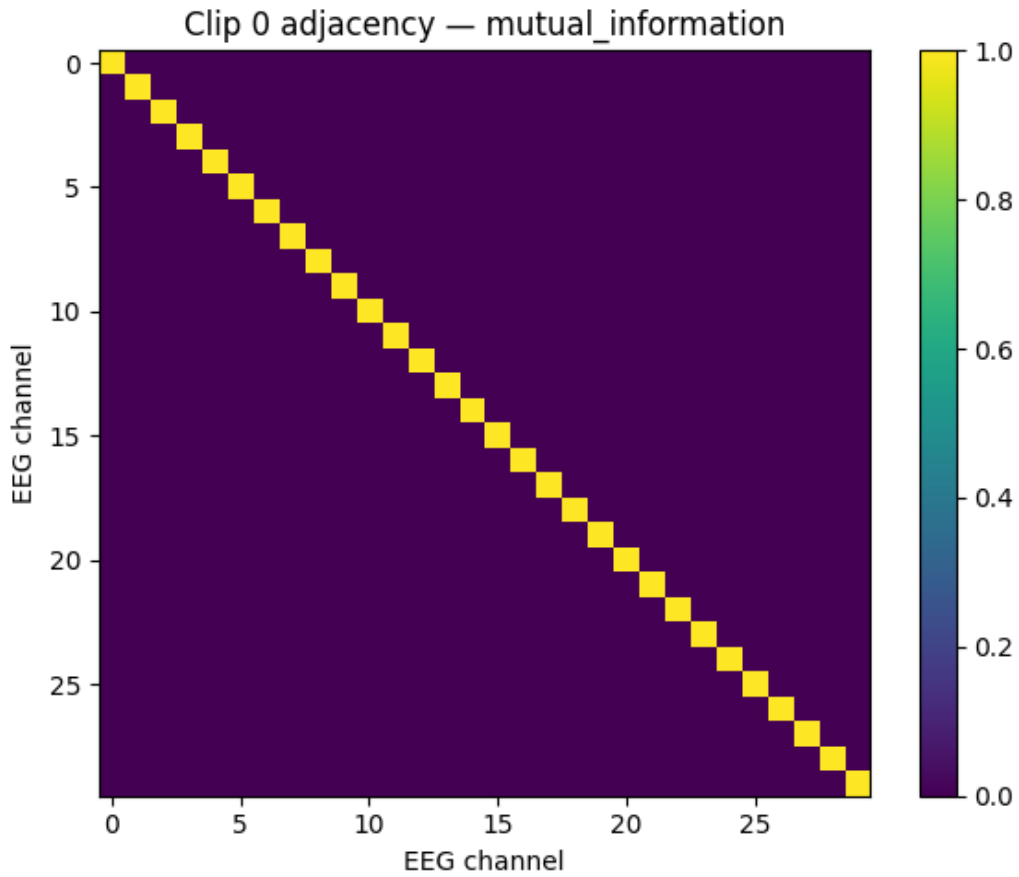


Fig. 5. Example clip-level mutual-information adjacency. Clip 0 retains only self-loops at the 0.70 threshold.

The example adjacency is a warning sign rather than an error by itself. The complete cache has 4.47% non-zero density, but some clips may reduce graph propagation to independent node processing. This should be audited before claiming that the performance gain comes from inter-channel graph learning.

13. Runtime, Checkpointing, and Reproducibility Controls

Runtime or control item	Verified value / status
Execution device	Tesla T4 with verified CUDA
Exact total wall-clock runtime	Not stored in the uploaded notebook outputs
Inner search workload	10 outer folds $\times$ 9 hyperparameter pairs $\times$ 3 inner folds
Final models	10 completed final-model checkpoints
Completed state at final-session start	274 completed runs and 9 active checkpoints detected
Selected final epochs	Mean 13.4; median 8.5; range 3–45
Checkpoint writing	Atomic last.pt files saved every epoch
Resume portability	CUDA-to-CUDA and CUDA-to-CPU loading through CPU mapping
Failure protection	Emergency bundle after Python errors or manual interruption
Final exports	All-checkpoints ZIP and complete resume-bundle ZIP

The checkpoint design is a strong operational contribution. It prevents completed inner folds from being retrained, preserves final checkpoints, and separates the paper-NCV and smoke-test profiles. However, reproducibility documentation should also record total training, validation, and test durations in the final result summary.

Conclusion

GNN4EEG provides a valuable large-scale benchmark for studying EEG graphs under both subject-dependent and subject-independent conditions. The uploaded main notebook closely follows the published cross-9 NCV protocol for DGCNN, RGNN, and SparseDGCNN, but its numerical results cannot be verified because the uploaded copy contains no stored outputs.

The completed HetEmotionNet-style notebook achieved $45.95 \pm 3.53\%$ mean outer-fold accuracy and $45.74 \pm 3.40\%$ macro-F1. Amusement was classified most reliably, while sadness was the weakest class. The result is reproducible from the notebook checkpoint outputs, but it is not an exact reproduction of the paper’s 30.8% HetEmotionNet NCV result because the feature source, adjacency construction, and model architecture differ materially.

The defensible final statement is: the GNN4EEG cross-9 evaluation pipeline was reconstructed, and a completed EEG-only HetEmotionNet-style reimplementation produced $45.95 \pm 3.53\%$ accuracy; strict paper-level replication remains incomplete until the official DE/LDS input, official HetEmotionNet implementation, sample-specific adjacency procedure, and verifiable outputs for all four models are run under a controlled configuration.

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